

21	A	Electric field strength $E = \frac{Q}{4\pi\epsilon_0 r^2}$. Since a graph of E vs $\frac{1}{r^2}$ is plotted, gradient = $\frac{Q}{4\pi\epsilon_0}$. Hence $Q = 4\pi\epsilon_0 \times$ gradient
22	B	Since the heating elements are in parallel they have the same p.d. across them